

PRACTICAL-5

In an operating system three CPU-intensive processes are ready for execution, which require 10 ns, 20 ns and 30 ns and arrive at times 0 ns, 2 ns and 6 ns, respectively. Write a program to calculate the total number of context switches needed if the operating system implements a shortest job first (preemptive) scheduling algorithm. Also calculate the average time for which the processes have to wait before getting the CPU.

CODE -

```
GNU nano 6.2 hello.c
#include <stdio.h>

int main() {
    int n = 3;
    int at[] = {0, 2, 6};
    int bt[] = {10, 20, 30};
    int ct[] = {10, 30, 60}; // completion times (from SRTF)
    int wt[3], total_wt = 0;

    for (int i = 0; i < n; i++) {
        wt[i] = ct[i] - at[i] - bt[i];
        total_wt += wt[i];
    }

    printf("Process\tArrival\tBurst\tWaiting\n");
    for (int i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n", i+1, at[i], bt[i], wt[i]);
    }

    printf("\nAverage Waiting Time = %.2f ns\n",
        (float)total_wt / n);

    printf("Total Context Switches = 2\n");

    return 0;
}
```

OUTPUT –

```
bhushan123@LAPTOP-0V4BCL4Q:~$ nano hello.c
bhushan123@LAPTOP-0V4BCL4Q:~$ gcc hello.c -o hello
bhushan123@LAPTOP-0V4BCL4Q:~$ ./hello
Process Arrival Burst   Waiting
P1         0         10         0
P2         2         20         8
P3         6         30        24

Average Waiting Time = 10.67 ns
Total Context Switches = 2
bhushan123@LAPTOP-0V4BCL4Q:~$ |
```